

Basics in Statistics

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Looking Beyond the Basics



6 October 2026, Afternoon

Outline

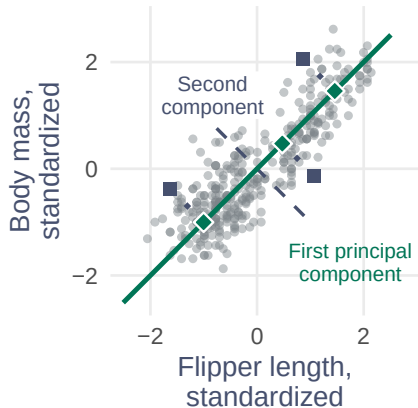
Principal component analysis for big data sets

- The idea in two dimensions
- Correlated measurements
- Standardizing before PCA
- Scores and loadings
- How many components?
- Reading a biplot
- Limitations of PCA

Decision trees: A popular prediction tool from machine learning

Writing professional statistical reports with Quarto

The idea of principal component analysis



Flipper length and body mass of 342 penguins, both standardized as z-scores (Unit 1); correlation 0.87. The squares mark three birds, the diamonds their positions on the green line.

Definition (First principal component, score)

The **first principal component** is the direction in which the points spread out most. The **score** of a study unit is its position along this direction.

- This one score per bird keeps 94 % of the variation in the two measurements. A high score means a large bird.
- The second component keeps the other 6 %.

Principal component analysis

Example (Correlation matrix of four measurements on 342 penguins)

	BL	BD	FL	BM
bill length (BL)	1			
bill depth (BD)	-0.24	1		
flipper length (FL)	0.66	-0.58	1	
body mass (BM)	0.60	-0.47	0.87	1

Definition (Principal component analysis, PCA)

PCA turns correlated numerical measurements into the same number of new variables, the **principal components**, each a weighted sum of the standardized measurements. The first component spreads the study units out most, the second next most, and so on. The components are uncorrelated; the first few summarize the data.

The same works for hundreds of columns, such as expression levels of many genes.

Standardizing before PCA

Definition (Standardization)

A column is **standardized** by turning every value into its z-score: Subtract the column mean, then divide by the column standard deviation. Afterwards every column has mean 0 and variance 1.

Example (Two PCAs of the 342 penguins: Share and weights of component 1)

analysis	share	bill length	bill depth	flipper length	body mass
unscaled	99.99 %	0.004	-0.001	0.015	1.000
standardized	68.8 %	0.455	-0.400	0.576	0.548

- Unscaled, the column with the largest numbers takes over: Body mass in grams has a variance of 643 131 g², the others below 200.
- In R: `prcomp(data, scale. = TRUE)`.

Scores and loadings

Definition (Loadings and scores)

The weights of a component are its **loadings**. The **score** of a study unit, its position along the component, is the weighted sum of its standardized measurements.

Example (First component of the standardized penguin measurements)

	bill length	bill depth	flipper length	body mass	score
loadings	0.455	-0.400	0.576	0.548	
Adelie penguin	-1.10	0.73	-2.06	-1.31	-2.70
Gentoo penguin	2.87	-0.08	2.07	2.30	3.79

Adelie: $0.455 \cdot (-1.10) - 0.400 \cdot 0.73 + 0.576 \cdot (-2.06) + 0.548 \cdot (-1.31) = -2.70$

A study unit scores high if its values are large where the loadings are positive and small where they are negative. R may reverse all signs; the component is the same.

Exercise: Loadings and scores

Exercise 4.1 (5 minutes, in pairs)

Loadings of the first two components of the four standardized penguin measurements:

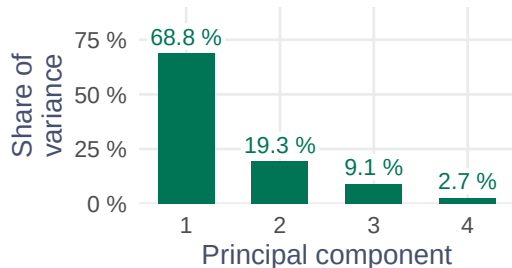
	bill length	bill depth	flipper length	body mass
component 1	0.455	-0.400	0.576	0.548
component 2	-0.597	-0.798	-0.002	-0.084

1. What does a high score on component 1 say about a penguin?
2. What does component 2 measure?
3. A Chinstrap penguin scores 0.19 on component 1 and -2.61 on component 2. Describe this bird in words.

How many components?

Definition (Proportion of variance explained)

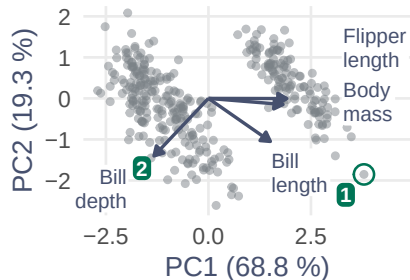
Each component carries a share of the total variance; all shares add up to 100 %. The cumulative share of the first components is the sum of their shares.



Scree plot of the PCA of the four standardized penguin measurements.

- Keep the components before the bars flatten out, here 2; components 3 and 4 add only 9.1 % and 2.7 %.
- Or keep the fewest components whose cumulative share reaches, say, 80 %; here two reach 88.2 %.
- Report which rule you used.

The biplot



342 penguins and four standardized measurements; PC1 and PC2 are components 1 and 2. Bird (1) and the arrow of bill depth (2) are marked.

Definition (Biplot)

A **biplot** shows the scores on the first two components as points and the loadings as arrows. In R: `biplot(pca)`.

- A point is a study unit, and nearby points are similar. An arrow is a variable and points to where it is large.
- Arrows in the same direction mean a positive correlation, opposite arrows a negative one, and a right angle almost none. The plot shows two of four dimensions, so this is only a rough guide.

Limits of PCA

- PCA is **exploratory**: It gives no p-value, no standard error, and no test. A pattern in the display is a hypothesis for the next study.
- PCA needs numerical, correlated columns. If the columns are uncorrelated, every component carries about the same share, and nothing is summarized.
- PCA is **unsupervised**: It uses no outcome and no group labels. A direction of large variance need not separate the groups.

Exercise: PCA in R

Go to `u4_beyond_basics.R`, section 4.1

R exercise 4.1 (15 minutes, in pairs)

Data: `iris`, four flower measurements in cm, $n = 150$.

1. Work through section 4.1: Run every demo, then solve the task below it for `iris`.
2. Is a PCA worth doing on these four columns? Why?
3. Report the share of the first component with and without scaling. All columns are in centimetres: Does scaling matter?
4. How many components do you need for 80 %?

Outline

Principal component analysis for big data sets

Decision trees: A popular prediction tool from machine learning

- Predicting instead of explaining

- Recursive partitioning

- Overfitting

- Test data and confusion matrix

Writing professional statistical reports with Quarto

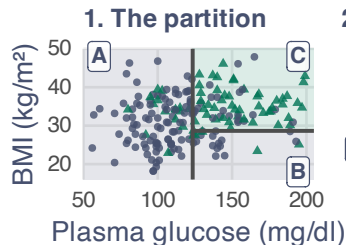
Explanation and prediction

Definition (Supervised and unsupervised methods)

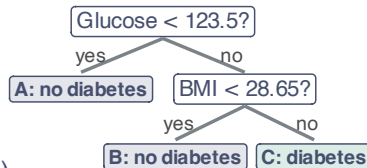
A method is **supervised** if it learns from an observed outcome, such as diabetes yes or no, and predicts it from the other variables (the predictors). PCA is **unsupervised**: It uses no outcome.

	explaining (Units 1 to 3)	predicting (this section)
aim	estimate a parameter and its uncertainty	predict the outcome of a new study unit
quality	standard error, confidence interval, p-value	error on new data that were not used for fitting

Decision trees



2. The same rule, as a tree



The 200 women of Pima heritage in MASS::Pima.tr: Triangles mark the 68 with diabetes, circles the 132 without. The two lines split them into the groups A, B, and C.

Definition (Decision tree)

A **decision tree** predicts the outcome by a sequence of yes/no questions such as "Is glucose below 123.5?". Each question is chosen so that the two groups it creates are as **pure** as possible, that is, nearly everyone in a group has the same outcome. A new study unit follows the answers down to a **leaf**, which carries the prediction.

Example: A classification tree in R

```
—— rpart(type ~ glu + bmi, data = Pima.tr, maxdepth = 2) ——  
node), split, n, loss, yval, (yprob)  
1) root 200 68 No (0.6600000 0.3400000)  
  2) glu< 123.5 109 15 No (0.8623853 0.1376147) *  
  3) glu>=123.5 91 38 Yes (0.4175824 0.5824176)  
    6) bmi< 28.65 21  5 No (0.7619048 0.2380952) *  
    7) bmi>=28.65 70 22 Yes (0.3142857 0.6857143) *
```

Each line is a node: Its number, the split, the number of training women, how many of them the tree misclassifies (loss), the predicted class, and the proportions of No and Yes. A star marks a leaf; the leaves 2, 6, and 7 are A, B, and C in the figure.

- Glucose below 123.5 mg/dl: “no diabetes”. The tree misclassifies 15 of these 109 women.
- Glucose at least 123.5 mg/dl and BMI at least 28.65: “diabetes”, with the estimated probability $48/70 = 0.69$.

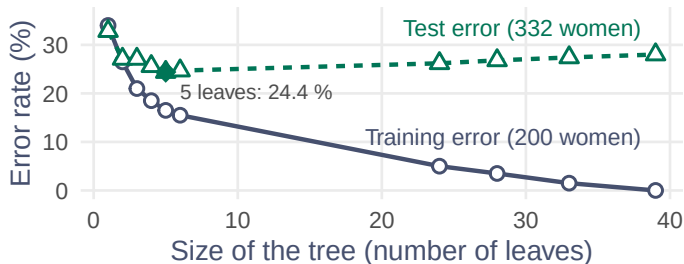
Exercise: Classifying with a printed tree

```
      rpart(type ~ glu + bmi, data = Pima.tr, maxdepth = 2)
node), split, n, loss, yval, (yprob)
1) root 200 68 No (0.6600000 0.3400000)
  2) glu< 123.5 109 15 No (0.8623853 0.1376147) *
    3) glu>=123.5 91 38 Yes (0.4175824 0.5824176)
      6) bmi< 28.65 21 5 No (0.7619048 0.2380952) *
        7) bmi>=28.65 70 22 Yes (0.3142857 0.6857143) *
```

Exercise 4.2 (5 minutes, in pairs)

1. A woman has a plasma glucose of 150 mg/dl and a BMI of 32. What does the tree predict, and with which estimated probability?
2. The same for a woman with glucose 110 mg/dl and BMI 40.
3. How many leaves does it have, and how many training women does it misclassify?
4. Which leaf would you trust least? Look at how many women each leaf rests on.

Overfitting



Trees with all seven predictors and 1 to 39 leaves, fitted to the 200 training women (Pima.tr); error on these women and on 332 other women (Pima.te).

Definition (Overfitting)

A model **overfits** if it learns the chance patterns of the training data: Its training error is small, but on new data it does worse than a simpler model.

Test error and confusion matrix

Definition (Confusion matrix)

The **confusion matrix** is the table of predicted against true classes in the test data. Correct predictions are on the diagonal; the other cells show *which* errors are made.

```
the example tree on Pima.te
      true
predicted No Yes
      No  183  44
      Yes   40  65
```

The test error is $(44 + 40)/332 = 25\%$; always predicting “No” would give $109/332 = 33\%$.

The tree finds 65 of the 109 women with diabetes (sensitivity 60%) and correctly predicts “No” for 183 of the 223 without (specificity 82%).

Exercise: Decision trees in R

Go to `u4_beyond_basics.R`, section 4.2

R exercise 4.2 (12 minutes, in pairs)

Data: `MASS::Pima.tr` (training) and `MASS::Pima.te` (test), outcome type (diabetes), seven predictors.

1. Work through section 4.2: Run every demo, then solve the task below it.
2. Fit a tree with all seven predictors. Which variables does it use?
3. Compute its training and test error with `err()`. Is its test error lower than that of the tree with glucose and BMI only?
4. Print the confusion matrix of your tree on the test data with `table()`. Sensitivity and specificity?

Outline

Principal component analysis for big data sets

Decision trees: A popular prediction tool from machine learning

Writing professional statistical reports with Quarto

- Reproducibility

- How R, Quarto, and the rendered document fit together

- Project organization

- Reporting transparently

Reproducibility

Definition (Reproducible and replicable)

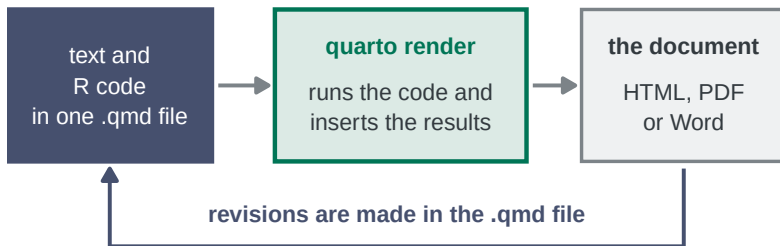
An analysis is **reproducible** if the same data and the same code give the same result, on any computer and at any later time. A finding is **replicable** if new data on the same question lead to the same conclusion.

Example: The first line of the script behind a published figure.

```
weights <- read.csv("/Users/anna/Desktop/Paper_final_v3/weights.csv")
```

- The line fails on a new laptop, on the computer of a co-author, and after the folder has been renamed: An absolute path names a location on one computer.
- Reasons to work reproducibly: A reviewer asks for one more model, corrected data change every number, somebody else takes over, or after six months nobody remembers which model produced Table 2.

R, Quarto, and the rendered document



Definition (Quarto document)

A **Quarto document** is a text file with the extension `.qmd`. It holds a header with settings, the report text, and the R code in **code chunks**. `quarto render` runs the code in a new R session and puts text and results together into one document.

Code chunks and inline code

```
```${r}
#| label: fig-scatter
#| fig-cap: "Body mass against flipper length. Each point is one bird."
ggplot(penguins_complete,
 aes(flipper_length_mm, body_mass_g, shape = species, colour = species)) +
 geom_point(alpha = 0.75, size = 1.8)
```
```

Of the `\${r} n\_all` birds in the data set, `\${r} n\_used` have complete measurements and are used below; `\${r} n\_all - n\_used` are dropped.

- A code chunk starts with `\${r}` and ends with ```. Lines that begin with #| set chunk options, here the label and the figure caption. A label that starts with fig- can be cited: @fig-scatter is rendered as “Figure 1”.
- **Inline code** of the form `\${r} expression` is evaluated during rendering and replaced by its value: *Of the 344 birds in the data set, 342 have complete measurements and are used below; 2 are dropped.* No number is typed by hand.

Project organization

```
my_project/  
  my_project.Rproj  
  data/  
    raw/  
    processed/  
  R/  
    01_clean.R  
    02_analyse.R  
  figures/  
  report.qmd  
  README.md
```

1. All paths are relative to the project root, the folder with the `.Rproj` file: `"data/raw/weights.csv"`. No `setwd()`.
2. `data/raw/` is read-only. Wrong values are corrected in code, where the correction stays visible.
3. Scripts are numbered in the order they run and start with `library()` and `set.seed()`.

Example: A minimal reproducible report

The file `minimal_report.qmd` fits

`lm(body_mass_g ~ flipper_length_mm + species)` to the penguins. It contains two tables and a figure with captions, a section “What this report does not claim”, and `sessionInfo()` at the end.

Example (Result sentence of the rendered report; every number is inline code)

Within a species, a flipper that is one millimetre longer goes with a body mass that is on average 40.7 g larger (95 % CI 34.7 to 46.7 g). The model reproduces 78 % of the variation in body mass.

- The report uses 342 of the 344 birds and says why: Two rows lack flipper length or body mass.
- It gives the size of the effect with an interval, in the units of the measurement, not a p-value alone.

Exercise: Rendering a report

Exercise 4.3 (12 minutes, in pairs)

1. Download `minimal_report.qmd` with the button “Download Quarto” of Unit 4 on the course page, open it in RStudio, and press “Render”.
2. Find in the source the inline code that produces the sentence about the slope, and the chunk options that produce the figure caption.
3. Exclude the Gentoo penguins: In the chunk data, add `species != "Gentoo"` as a third condition inside `filter()`, after a comma. Render again.
4. Which numbers, tables, and figures of the report have changed? Compare the slope, its confidence interval, and R^2 with the first rendering: 40.7 g, 34.7 to 46.7 g, and 78 %.

Wrap-up

- Unit 1: Describing data, probability, and distributions
- Unit 2: Estimation, confidence intervals, and tests
- Unit 3: Linear and logistic regression
- Unit 4: PCA, decision trees, and reproducible reports

Slides, R scripts, and solutions: loelschlaeger.de/basics_in_statistics

Thank you 🙏